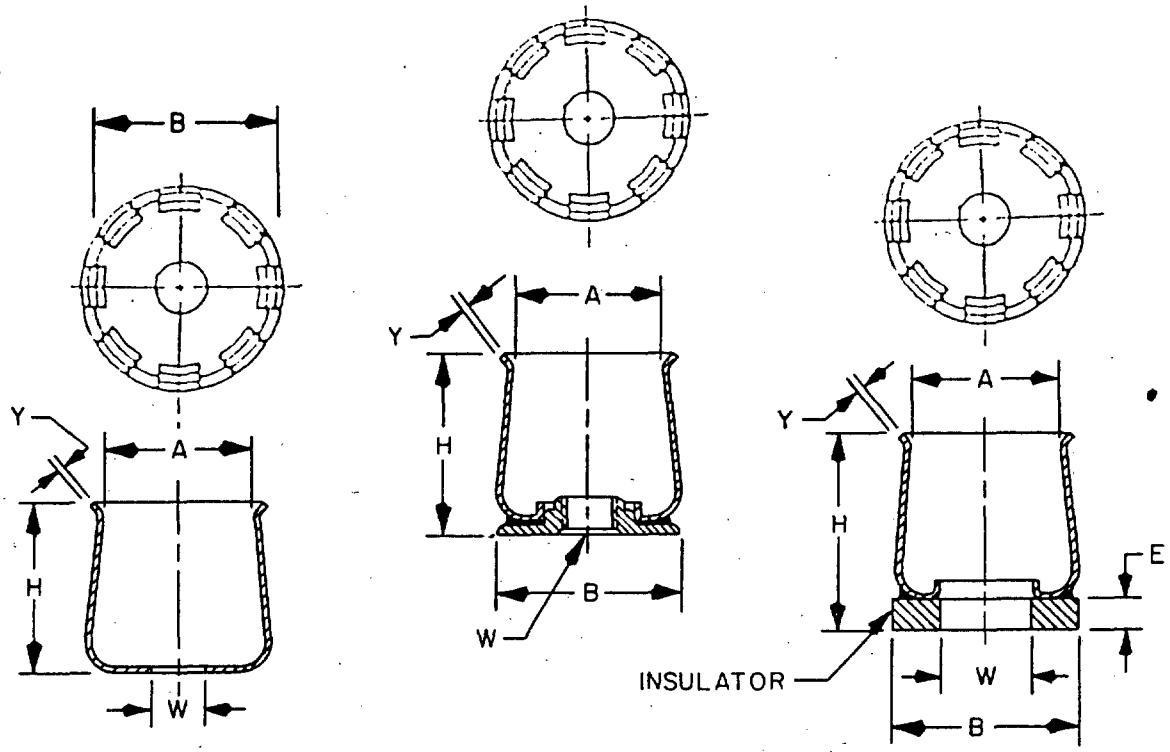
MILITARY SPECIFICATION SHEET

HEAT SINKS, SEMICONDUCTOR DEVICES, CLASSES 2, 3, AND 5 (RETAINER CLIP TYPE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-H-87111.

Part or identifying number (PINs) M87111/02-3*07 through M87111/02-3*15 are inactive for new design as of 26 December 1988.

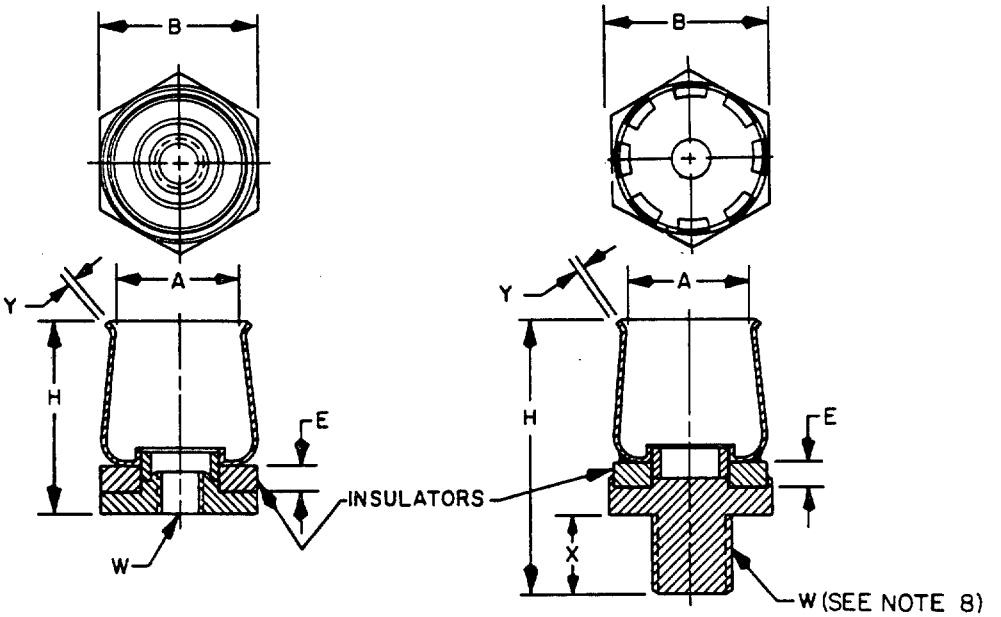


CONFIGURATION A CONFIGURATION B CONFIGURATION C

FIGURE 1. Heat sinks (retainer clip type).

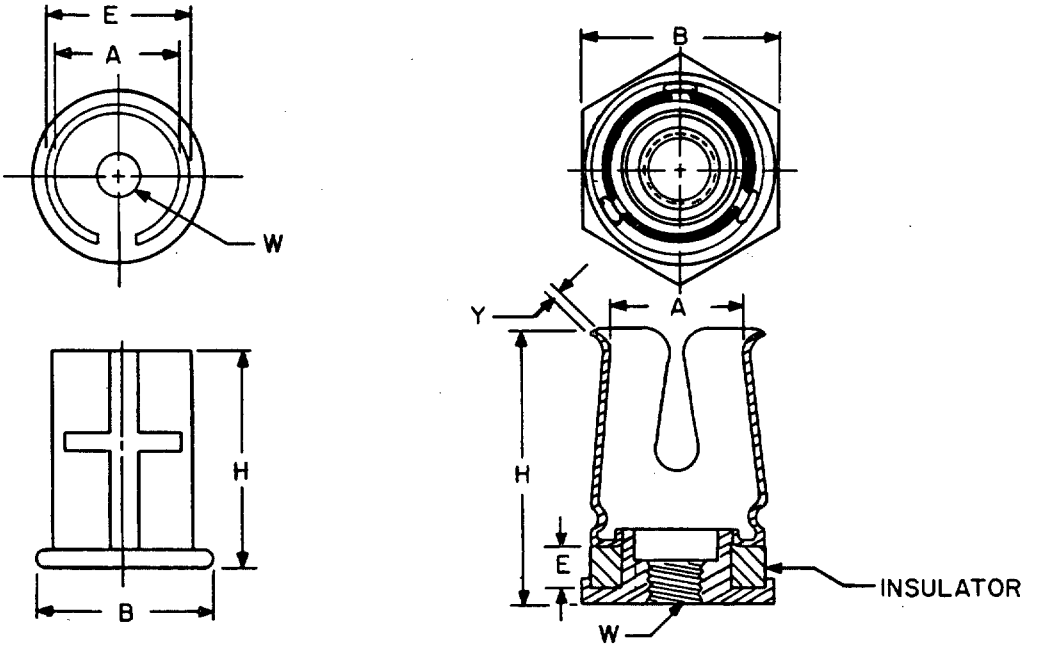
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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CONFIGURATION D

CONFIGURATION E

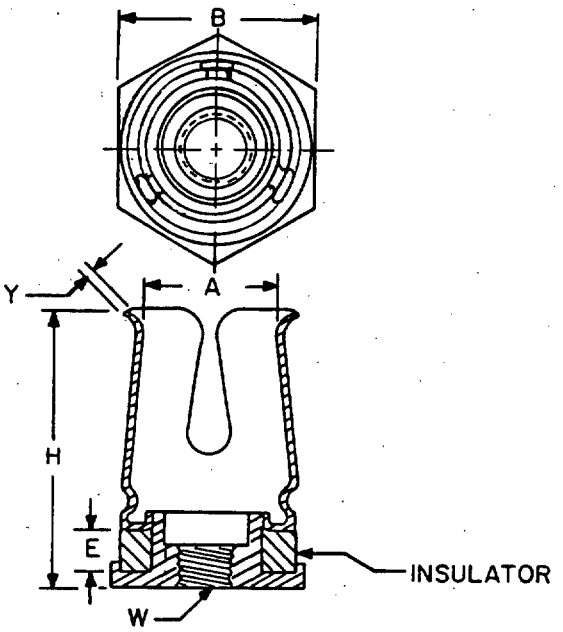


CONFIGURATION F

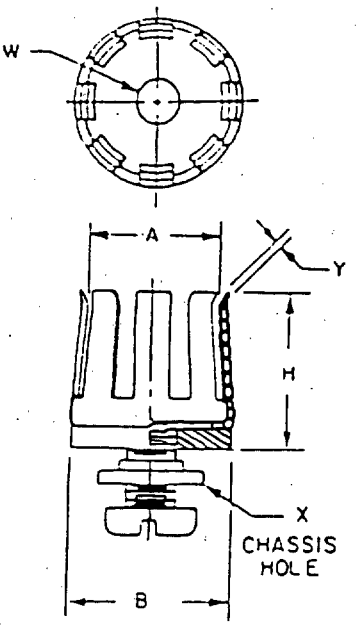
CONFIGURATION G

FIGURE 1. Heat sinks (retainer clip type) - Continued.

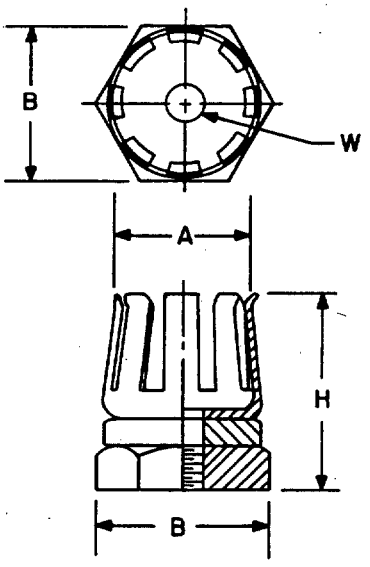
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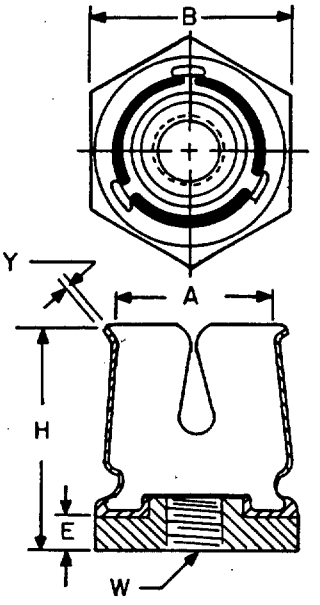
CONFIGURATION H



CONFIGURATION I



CONFIGURATION J



CONFIGURATION K

FIGURE 1. Heat sinks (retainer clip type) - Continued.

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Dash number (see note 7)	JEDEC case size	Config-uration	Dimensions (see notes)						
			A Dia +.005 (0.13)	B	E	H	W	X	Y
2*01	TO-18	A	.170 -.010 (4.32 - 0.25)	.250 ±.015 (6.35 ±0.38)	N/A	.219 ±.015 (5.56 ±0.38)	.067 +.004 -.001 (1.70 +0.10 -0.03)	N/A	.012 (0.30)
2*02	TO-5	A	.295 -.015 (7.49 -0.38)	.375 ±.010 (9.52 ±0.25)	N/A	.320 ±.035 (8.13 ±0.89)	.098 +.015 -.001 (2.49 +0.38 -0.03)	N/A	.013 (.33)
2*03	TO-8	A	.470 -.015 (11.94 -0.38)	.545 ±.015 (13.84 ±0.38)	N/A	.281 ±.015 (7.14 ±0.38)	.130 (3.30)	N/A	.010 (.25)
2*04	TO-18	B	.170 -.010 (4.32 -0.25)	.235 ±.025 (5.97 ±0.64)	N/A	.290 ±.025 (7.37 ±0.64)	.086-56 UNC-2B	N/A	.012 (.31)
2*05	TO-5	B	.295 -.015 (7.49 -0.38)	.375 ±.015 (9.52 ±.038)	N/A	.370 ±.035 (9.40 ±0.38)	.112-40 UNC-2B	N/A	.013 (.33)
2*06	TO-8	B	.475 -.015 (12.07 -0.38)	.545 ±.015 (13.84 ±0.38)	N/A	.370 ±.015 (9.40 ±0.38)	.112-40 UNC-2B	N/A	.010 (.25)
3*07	TO-18	C	.176 -.015 (4.47 -0.38)	.220 ±.015 (5.59 ±0.38)	.060 (1.52)	.281 (7.14)	.067 (1.70)	N/A	.012 (.31)
3*08	TO-5	C	.305 -.015 (7.75 -0.38)	.375 ±.015 (9.52 ±0.38)	.060 (1.52)	.385 ±.015 (9.78 ±0.38)	.098 (2.49)	N/A	.013 (.33)
3*09	TO-8	C	.470 -.015 (11.94 -0.38)	.545 ±.015 (13.84 ±0.38)	.070 (1.78)	.281 ±.015 (7.14 ±0.38)	.130 (3.30)	N/A	.010 (.25)
3*10	TO-8	D	.470 -.015 (11.94 -0.38)	.545 ±.015 (13.84 ±0.38)	.070 (1.78)	.281 ±.015 (7.14 ±0.38)	.112-40-UNC-2A	N/A	.010 (.25)
3*11	TO-18	E	.170 -.010 (4.32 -0.25)	.250 ±.010 (6.35 ±0.25)	.060 (1.52)	.550 ±.015 (13.97 ±0.38)	.112-40-UNC-2A	.190 (4.83)	.012 (.31)
3*12	TO-5	E	.295 -.015 (7.49 -0.38)	.375 ±.010 (9.52 ±0.25)	.060 (1.52)	.656 +.012 -.050 (16.66 +0.30 -1.27)	.138-32 UNC-2A	.190 (4.83)	.013 (.33)

FIGURE 1. Heat sinks (retainer clip type) - Continued.

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Dash number (see note 7)	JEDEC case size	Con-figu-ration	Dimensions (see notes)						
			A Dia +.005 (0.13)	B	E	H	W	X	Y
3*13	TO-5	F	.315 -.006 (8.00 -0.15)	.375 ±.007 (9.52 ±0.18)	.351 (8.92)	.400 ±.016 (10.16 ±0.41)	.112-40-UNC-2B	N/A	N/A
3*14	TO-18	G	.170 -.010 (4.32 -0.25)	.250 ±.015 (6.35 ±0.38)	.050 (1.27)	.350 ±.015 (8.89 ±0.38)	.086-56-UNC-2B	N/A	.010 (0.25)
3*15	TO-5	H	.295 -.015 (7.49 -0.38)	.375 ±.015 (9.52 ±0.38)	.050 (1.27)	.445 ±.015 (11.30 ±0.38)	.112-40-UNC-2B	N/A	.010 (0.25)
2*16	TO-5	I	.295 -.015 (7.49 -0.38)	.375 ±.015 (9.52 ±0.38)	N/A	.370 ±.015 (9.40 ±0.38)	.112-40-UNC-2B	.156 (3.96)	.013 (0.33)
2*17	TO-5	J	.295 -.015 (7.49 -0.38)	.375 ±.015 (9.52 ±0.38)	N/A	.410 ±.015 (10.41 ±0.38)	.112-40-UNC-2B	N/A	.013 (0.33)
5C18	TO-18	J	.170 -.010 (4.32 -0.25)	.250 ±.015 (6.35 ±0.38)	N/A	.290 ±.015 (7.37 ±0.38)	.086-56-UNC-2B	N/A	.010 (0.25)
5C19	TO-18	E	.170 -.010 (4.32 -0.25)	.250 ±.010 (6.35 ±0.25)	N/A	.495 ±.015 (12.57 ±0.38)	.086-56-UNC-2A	.187 (4.75)	.010 (0.25)
5C20	TO-5	E	.295 -.010 (7.49 -0.25)	.375 ±.015 (9.52 ±0.38)	N/A	.585 ±.020 (14.86 ±0.46)	.138-32-UNC-2A	.187 (4.75)	.010 (0.25)
2*21	TO-18	I	.170 -.010 (4.32 -0.25)	.215 ±.015 (5.46 ±0.38)	N/A	.370 ±.015 (9.40 ±0.38)	.086-56-UNC-2B	.125 (3.17)	.012 (0.30)
2*22	TO-8	I	.470 -.015 (11.94 -0.38)	.550 ±.015 (13.97 ±0.38)	N/A	.370 ±.015 (9.40 ±0.38)	.112-40-UNC-2B	.156 (3.96)	.010 (0.25)
5C23	TO-18	K	.170 -.010 (4.32 -0.25)	.250 ±.015 (6.35 ±0.38)	.050 (1.27)	.295 ±.015 (7.49 ±0.38)	.086-56-UNC-2B	N/A	.010 (0.25)
5C24	TO-5	K	.295 -.010 (7.49 -0.25)	.375 ±.015 (9.52 ±0.38)	.090 (2.29)	.400 ±.015 (10.16 ±0.38)	.112-40-UNC-2B	N/A	.010 (0.25)
5C25	TO-8	K	.470 -.010 (11.94 -0.25)	.550 ±.015 (13.97 ±0.38)	N/A	.375 ±.015 (9.52 ±0.38)	.112-40-UNC-2B	.156 (3.96)	N/A
5C26	TO-18	B	.170 -.010 (4.32 -0.25)	.215 ±.015 (5.46 ±0.38)	N/A	.330 ±.015 (8.38 ±0.38)	.086-56-UNC-2B	N/A	.010 (0.25)
5C27	TO-5	B	.295 -.010 (7.49 -0.25)	.365 ±.015 (9.27 ±0.38)	N/A	.365 ±.010 (9.27 ±0.38)	.112-40-UNC-2B	N/A	.010 (0.25)
2C28	TO-18	E	.170 -.010 (4.32 -0.25)	.250 ±.010 (6.35 ±0.25)	N/A	.495 ±.020 (12.57 ±0.51)	.138-32-UNC-2A	.187 (4.75)	.012 (0.30)
2C29	TO-18	E	.170 -.010 (4.32 -0.25)	.250 ±.010 (6.35 ±0.25)	N/A	.495 ±.020 (12.57 ±0.51)	.112-40-UNC-2A	.187 (4.75)	.012 (0.30)

FIGURE 1. Heat sinks (retainer clip type) - Continued.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
5. Drawing depicts only general design characteristics and is not intended to limit specific manufacturing designs.
6. Number of fingers may vary depending on manufacturer.
7. The asterisk in the dash number represents the finish (i.e., B, C, D, G, or N).
8. .0625 inch (1.588 mm) undercut to minimum thread diameter at top portion of threaded stud.

FIGURE 1. Heat sinks (retainer clip type) - Continued.

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REQUIREMENTS:

Dimensions and configurations: See figure 1.

Material options:
Beryllium copper (class 2).
Beryllium copper with beryllium oxide insulator (class 3).
Beryllium copper with epoxy insulator (class 5).
(The class is indicated in the PIN immediately following the dash).

CAUTION: Class 3 devices contain beryllium oxide ceramics. Heat sink-insulators of this type are not hazardous in their finished state; however, grinding, sandblasting, machining, sectioning, or similar operations should not be performed on these parts unless proper safety precautions are observed. These operations may produce beryllium dust or fumes which, if inhaled, could cause serious pulmonary disease. Inhalation of beryllium particles in excess of OSHA occupational exposure standards should be avoided. Class 3 parts may be recycled or disposed of according to the manufacturer's instructions.

Finish options:
Black chemical (finish B).
Insulating coating (finish C).
Cadmium plating (finish D).
Gold plating (finish G).
Nickel plating (finish N).

Mounting surface flatness: Mounting surfaces shall be flat within .003 inch (0.08 mm) per inch of Total Indicator Reading (TIR).

Surface roughness: Not applicable.

JEDEC case sizes: See figure 1.

Thermal resistance:

a. TO-5: 15°C/watt maximum between 1-watt and 3-watt dissipation.

b. TO-8: 12°C/watt maximum between 1.5-watt and 12-watt dissipation.

c. TO-18: 15°C/watt maximum between 0.5-watt and 2-watt dissipation.

PIN: M87111/02- (plus dash number from figure 1). Complete PIN example: M87111/02-2B01.

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TABLE I. Supersession data.

Superseding PIN in accordance with MIL-H-87111/2C 26 December 1988	Superseded PIN 1/ in accordance with MIL-H-87111/2B 18 April 1988
M87111/02-5C26	M87111/02-3*07
M87111/02-5C27	M87111/02-3*08
M87111/02-5C25	M87111/02-3*09
M87111/02-5C25	M87111/02-3*10
M87111/02-5C19	M87111/02-3*11
M87111/02-5C20	M87111/02-3*12
M87111/02-5C24	M87111/02-3*13
M87111/02-5C23	M87111/02-3*14
M87111/02-5C24	M87111/02-3*15

1/ The asterisk in the PIN represents the finish (i.e., B, C, D, G, or N).

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - TD
Air Force - 11, 19, 99
DLA - ES

User activities:
Navy - AS, CG, MC, SH

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5999-0275)

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